Caleb Strout

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Education

Vanderbilt University Nashville, TN

Bachelor of Engineering in Mechanical Engineering (Concentration in Aerospace) *May 2027*

- GPA: 3.91, Dean's List (all semesters)
- Relevant Coursework: Aerospace Propulsion, Heat Transfer, Fluid Mechanics, Thermodynamics, Mechanical Design, Dynamics, Statics, System Dynamics, Mechanics of Materials, Machine Analysis and Design

Technical Skills

Programming Languages: Python, MATLAB, LabVIEW, R, C++
Tools: Excel, CATIA, Creo, SOLIDWORKS CAD/CAM, Siemens NX, OnShape, AutoCAD, Ansys, Arduino, 3-D Printing, Raspberry PI, CNC, Milling

Experience

General Atomics June 2026 – August 2026

Fuel Systems Engineer Intern *Adelanto, CA*

- Conducted manifold testing on the Mojave STOL system, using CFD analysis to ensure compatibility across multiple solenoid valve configurations.
- Designed test setup of Secon Heat Exchanger for Mojave STOL on engine test stand to be done in October 2026.
- Analyzed YFQ-42A flight data to increase pitch and roll degree range for fuel mismatch to ensure safety with leakages and CG balance.
- Created Excel calculator to investigate fuel time, overpressure safety factors and active CG for Mojave STOL single point refuel process.

Whisper Aero Feb 2026 – May 2026

Test Engineer Intern *Nashville, TN*

- Designed and assembled spin-to-burst test chamber to evaluate failure in the bladed disk of a propulsor scaled for a leaf blower product, stepping from 20,000 rpm to 35,000 rpm.
- Prototyped and validated a stubby nozzle attachment through blow force, acoustic, and balancing tests to ensure more concentrated air flow as fluid exits the duct of the leaf blower, resulting in an intensive sound of 50 db(A) at 50 feet away and a maximum blow force of 30 N within product specifications.

Vanderbilt Aerospace Design Laboratory June 2025 – August 2025

Undergraduate Research Assistant *Nashville, TN*

- Engineered gear system to deploy fins simultaneously with one servo motor, inducing different drag forces dependent on the fins' angular positions.
- Integrated a VN-100 IMU with a Raspberry Pi 4 using Python to collect apogee related data to determine when fins should be deployed with 95% accuracy.
- Introduced induced drag to a rocket trajectory analysis using MATLAB to model CD increase of rocket when fins deploy at a specific altitude, velocity, and time based on a predicted apogee with 95% accuracy based on test flight altitude and velocity data.
- Launched 68" rocket on 4 separate occasions with target apogees of 750 feet that successfully deployed fins and recorded apogees within 20 feet of the target.

Projects

NASA USLI Launch Competition | *SOLIDWORKS, R&D, C++, CNC, FEA, CFD* August 2025 – Present

- Validated mechanical design choices of vehicle and payload components, including payload extraction from vehicle, bolt tear-out, airframe loads, and coefficient of drag to ensure safety and reliability of design using Ansys FEA and CFD.
- Created a hardware-in-the-loop test bench to validate payload electronics and simulate rocket flight without motor usage using C++, Python, and TCP.
- Manufactured vehicle components, including fiberglass nosecones and carbon-fiber fins using composite layup and curing processes, as well as aluminum bulkheads using lathe machining and CNC operations.
- Tested payload design including shovel, latches, and DC motors to ensure system functionality.